

EXPERIMENT 25:

The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'Apriori' algorithm is chosen, and the command line is set to 'Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1'. The 'Start' button is pressed, and the 'Apriori output' is displayed. The output shows the minimum support (0.1), minimum metric (confidence), and the number of cycles performed (18). It also lists the generated sets of large itemsets and the best rules found.

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

21:24:32 - Apriori

Apriori

Minimum support: 0.1 (15 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 18

Generated sets of large itemsets:

Size of set of large itemsets L(1): 20
Size of set of large itemsets L(2): 15
Size of set of large itemsets L(3): 3

Best rules found:

1. petalwidth='(-inf-0.34]' 41 ==> class=Iris-setosa 41 <conf:(1)> lift:(3) lev:(0.18) [27] conv:(27.33)
2. petalwidth='(-inf-1.59]' 37 ==> class=Iris-setosa 37 <conf:(1)> lift:(3) lev:(0.16) [24] conv:(24.67)
3. petalwidth='(-inf-1.59]' petalwidth='(-inf-0.34]' 33 ==> class=Iris-setosa 33 <conf:(1)> lift:(3) lev:(0.15) [22] conv:(22)
4. petalwidth='(1.06-1.3]' 21 ==> class=Iris-versicolor 21 <conf:(1)> lift:(3) lev:(0.09) [14] conv:(14)
5. petalwidth='(5.13-5.72]' 18 ==> class=Iris-virginica 18 <conf:(1)> lift:(3) lev:(0.08) [12] conv:(12)
6. sepallength='(4.66-5.02]' petalwidth='(-inf-0.34]' 17 ==> class=Iris-setosa 17 <conf:(1)> lift:(3) lev:(0.08) [11] conv:(11.33)
7. sepallength='(2.96-3.2]' class=Iris-setosa 16 ==> petalwidth='(-inf-0.34]' 16 <conf:(1)> lift:(3.66) lev:(0.08) [11] conv:(11.63)
8. sepallength='(2.96-3.2]' petalwidth='(-inf-0.34]' 16 ==> class=Iris-setosa 16 <conf:(1)> lift:(3) lev:(0.07) [10] conv:(10.67)
9. petalwidth='(3.95-4.54]' 26 ==> class=Iris-versicolor 25 <conf:(0.96)> lift:(2.88) lev:(0.11) [16] conv:(8.67)
10. petalwidth='(1.78-2.02]' 23 ==> class=Iris-virginica 22 <conf:(0.96)> lift:(2.87) lev:(0.1) [14] conv:(7.67)

Status OK Log

The screenshot shows the Weka Explorer interface with the 'Apriori' algorithm results. A dialog box titled 'weka.gui.GenericObjectEditor' is open, displaying the configuration for the 'weka.associations.Apriori' class. The dialog box contains various settings such as 'car', 'classIndex', 'delta', 'doNotCheckCapabilities', 'lowerBoundMinSupport', 'metricType', 'minMetric', 'numRules', 'outputItemSets', 'removeAllMissingCols', 'significanceLevel', 'treatZeroAsMissing', 'upperBoundMinSupport', and 'verbose'.

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

21:24:32 - Apriori

Apriori

Minimum support: 0.1 (15 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 18

Generated sets of large itemsets:

Size of set of large itemsets L(1):
Size of set of large itemsets L(2):
Size of set of large itemsets L(3):

Best rules found:

1. petalwidth='(-inf-0.34]' 41 ==> class=Iris-setosa 41 <conf:(1)> lift:(3) lev:(0.18) [27] conv:(27.33)
2. petalwidth='(-inf-1.59]' 37 ==> class=Iris-setosa 37 <conf:(1)> lift:(3) lev:(0.16) [24] conv:(24.67)
3. petalwidth='(-inf-1.59]' petalwidth='(-inf-0.34]' 33 ==> class=Iris-setosa 33 <conf:(1)> lift:(3) lev:(0.15) [22] conv:(22)
4. petalwidth='(1.06-1.3]' 21 ==> class=Iris-versicolor 21 <conf:(1)> lift:(3) lev:(0.09) [14] conv:(14)
5. petalwidth='(5.13-5.72]' 18 ==> class=Iris-virginica 18 <conf:(1)> lift:(3) lev:(0.08) [12] conv:(12)
6. sepallength='(4.66-5.02]' petalwidth='(-inf-0.34]' 17 ==> class=Iris-setosa 17 <conf:(1)> lift:(3) lev:(0.08) [11] conv:(11.33)
7. sepallength='(2.96-3.2]' class=Iris-setosa 16 ==> petalwidth='(-inf-0.34]' 16 <conf:(1)> lift:(3.66) lev:(0.08) [11] conv:(11.63)
8. sepallength='(2.96-3.2]' petalwidth='(-inf-0.34]' 16 ==> class=Iris-setosa 16 <conf:(1)> lift:(3) lev:(0.07) [10] conv:(10.67)
9. petalwidth='(3.95-4.54]' 26 ==> class=Iris-versicolor 25 <conf:(0.96)> lift:(2.88) lev:(0.11) [16] conv:(8.67)
10. petalwidth='(1.78-2.02]' 23 ==> class=Iris-virginica 22 <conf:(0.96)> lift:(2.87) lev:(0.1) [14] conv:(7.67)

Status OK Log

weka.gui.GenericObjectEditor

weka.associations.Apriori

About

Class implementing an Apriori-type algorithm

More

Capabilities

car False

classIndex -1

delta 0.05

doNotCheckCapabilities False

lowerBoundMinSupport 0.1

metricType Confidence

minMetric 0.9

numRules 10

outputItemSets False

removeAllMissingCols False

significanceLevel -1.0

treatZeroAsMissing False

upperBoundMinSupport 1.0

verbose False

Open... Save... OK Cancel